

Docker & Containers for QA - Installation & Setup

Images, containers, Dockerfiles, volumes, ports and why containers help reproduce test environments.

Goal of this manual

This manual guides you through the complete installation and configuration process, from preparing the environment to final validation. Follow the steps in order and use each verification checkpoint to confirm that the setup was completed correctly.

Before you start

Confirm the supported operating system, runtime/dependencies and version requirements in the current official documentation before installing. The technical checklist below mirrors the validated tool-specific procedure.

Where to run commands

On Windows use PowerShell or Command Prompt; on macOS/Linux use Terminal. After changing PATH or environment variables, close and reopen the terminal before validating again.

Official references

<https://www.docker.com/>

Step-by-step installation and setup

1. Follow this tool-specific step: Baixe Docker Desktop/Engine da documentação oficial do seu sistema.
2. Follow this tool-specific step: Windows: instale Docker Desktop e mantenha WSL 2 quando recomendado pelo instalador/documentação.
3. Follow this tool-specific step: Inicie Docker e aguarde o engine ficar pronto.
4. Follow this tool-specific step: Abra novo terminal e execute `docker --version`.
5. Follow this tool-specific step: Execute `docker version` para validar cliente e servidor.
6. Follow this tool-specific step: Execute `docker run hello-world`.
7. Follow this tool-specific step: Confirme a mensagem de sucesso e remova o container/imagem depois se desejar.

Commands and validation

```
docker --version
```

```
docker version
```

```
docker run hello-world
```

First working use

Complete the first working use described in the steps above and confirm the expected tool-specific result before adding integrations or advanced configuration.

Common problems and fixes



Cannot connect to Docker daemon: inicie Docker Desktop/serviço; isso é diferente de PATH.

WSL/virtualização indisponível: corrija os requisitos do sistema antes de reinstalar.

Permissão no Linux: siga a configuração oficial do grupo docker, considerando implicações de segurança.

Completion criteria

Treat setup as complete when the version/execution checks work and the first use in this manual finishes without errors. Only then move to integrations or advanced configuration.